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Multi-Body Mooring System

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Abstract

As part of the engineering team responsible for evaluating vessel behaviour and motion, it is essential to assess critical offshore operational scenarios. One such scenario involves the mooring of a cargo barge to a construction vessel/barge operating in anchor mode, particularly during pipeline laying and structural lifting activities. This configuration affects the motion response of the construction vessel and reduces the allowable operational weather window. In other words, if the design excursions are limited to certain values for the operation purpose so weather limits to be reduced till fulfil the allowable excursions.

The analyses impacted by this setup are pipeline laying analysis, dynamic amplification factor (DAF) analysis, mooring analysis, and subsea installation analysis. The primary objective of this paper is to determine the optimal mooring system (from connector's mechanical properties point of view) between the cargo barge and the construction vessel that minimizes vessel motion under selected weather conditions. A multi-body mooring case study and its related conditions have been analysed and evaluated using MOSES software.

The optimum mooring configuration corresponds to the lowest motion response of the construction vessel, thereby enhancing workability limits and ensuring safety, reliability, and operational efficiency. The selection, arrangement, and design of fenders and soft mooring lines are critical in managing the interaction between the cargo barge and the construction vessel and act as the key to achieving an effective mooring solution.

Introduction

Enhancing the operational efficiency of offshore construction activities is a primary objective across all marine projects. However, achieving this goal often involves overcoming daily challenges during the simulation and planning phases of marine operations. One of the most common and critical operations is the mooring of a cargo barge (CB) to a construction vessel. While necessary, this configuration often acts as a limiting factor by reducing the vessel's allowable weather window, to illustrate the paper's purpose, a case

study on multi-body mooring is presented. Figure 1 is the Moses model for a multi-body mooring case study, the cargo barge Safeen B-281 moored to Construction vessel Safeen 3000 during pipeline laying operation.

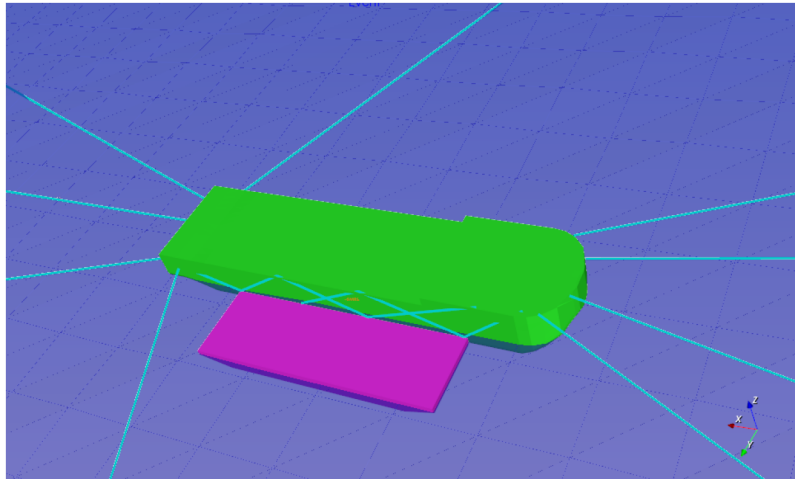


Figure 1—Moses Hydrodynamic model of multi-body mooring system case study during pipeline laying.

Modeling Input and Simulation Parameters

The hulls of all marine units involved in the analysis were modelled in MOSES [1] using over 1,000 pressure panels to ensure accurate hydrodynamic representation. The wave environment was defined using the JONSWAP spectrum with a peak enhancement factor (γ) of 5 [2], reflecting realistic offshore sea states for weather restricted operations.

All connectors were modelled based on manufacturer specifications, incorporating their respective elasticity curves to simulate realistic mechanical behaviour. Fenders and soft mooring lines were represented using non-linear elasticity models derived from performance charts, ensuring accurate force-displacement characteristics.

Pipeline tension was modelled as a constant $80 \text{ MTs} \pm 5 \text{ MTs}$ at the tensioners, reflecting operational variability. The use of pneumatic fenders resulted in more gradual and relaxed reaction forces compared to solid rubber fenders, which exhibit a sharp increase in force upon compression. This smoother response is a significant advantage in protecting the hulls of both the cargo barge and the construction vessel during mooring operations.

Marine Units Characteristics

Table 1 presents main principal data of Safeen3000 and CB Safeen B-281 [3][4].

Table 1—Marine units characteristics.

Data	Construction vessel (CV) Safeen 3000	Cargo barge (CB) Safeen B-281
Length between perpendicular (m) -LBP	144.2	86.01
Breadth (m)-B	37.8	27.45
Depth (m)-D	15.22	5.49
Draft (m) -during the analysis	5.34	3.8
Displacement (MT)	26694.82	7832.4
COG x,y,z (m) -center of gravity	78.38, 0, 9.1	41.98, 0, 6.5
COB x,y,z (m) -center of buoyancy	78.36, 0, 2.63	41.96, 0, 1.93
Gyration radius rx,ryy,rzz	11.7, 43.7, 44.8	8, 24, 24

Connectors

Table 2 presents main data of mooring lines, soft ropes and fenders.

Table 2—Connectors data

	Fenders	Mooring Lines	Soft Ropes
ID	F1, F2, F3	S1, S2, S3, S4, P1, P2, P3, P4	Sr1 to Sr7
Diameter (mm)	4500	64	48
Type	Pneumatic	Synthetic ropes	Synthetic ropes
MBL (MT)	-	282	179
Reaction force (MT)	Nonlinear as per Table 3	-	-

Table 3—Fender's reaction data verses deflection

Deflection (%), (m) of 4.5m diameter		Reaction force (kN)
0.0%	0.00	0.00
5.6%	0.25	9.58
11.1%	0.50	24.10
16.7%	0.75	46.23
22.2%	1.00	80.21
27.8%	1.25	130.65
33.3%	1.50	207.64

Cases

The conditions of multi-body mooring case study are categorized into two main categories, weather cases and mooring cases, accordingly all mooring cases are analyzed for all weather cases, the mooring cases are intact cases and damage cases.

Weather cases (w.Case). Wind and current are constant in all cases studied, wind speed is 30 knots, and current speed is 0.7 knots acting in same direction of waves, wave cases are as per Table 4

Table 4—Wave cases

Case no.	Hs (m)	Tp (Sec)	Direction (deg)
1	1.5	5	135
2	1.5	6	135
3	1.5	7	135
4	2	5	135
5	2	6	135
6	2	7	135
7	2.5	5	135
8	2.5	6	135
9	2.5	7	135

Mooring cases (m.Case). Intact cases are the initial case which is the case of CV without CB mooring and case0 which includes the CB mooring, the redundancy cases are case1 and case2. Damage mooring analysis is performed for the connectors between CV and CB, while the main mooring system of the CV is intact for all cases. Single line damage of main mooring lines isn't presented because the resultant excursion case can't conclude as a multi-body mooring effects. Table 5 presents mooring configuration of each study case

Table 5—Mooring cases

Case ID	Safeen 3000 (CV)	Safeen B-281 (CB)	Connectors used
Initial	Included in Simulation	Excluded from Simulation	Mooring lines and PL
Case0	Included in Simulation	Included in Simulation	All connectors active
Case1	Included in Simulation	Included in Simulation	F2, Sr3, Sr6 disconnected
Case2	Included in Simulation	Included in Simulation	F2, Sr2, Sr4, Sr7 disconnected

Analysis methodology and procedure

A comprehensive hydrodynamic time domain analysis was conducted to simulate the complex behavior of marine systems. We used the licensed MOSES software to perform these detailed analyses[1].3D-Diffraction analysis is used for 3 hours simulation [2], stiffness of the Pneumatic fenders and soft mooring lines are modeled, The study involved testing three distinct configurations to understand the variations in vessel motion under different conditions. Many iterations are analyzed with different mooring characteristics and configurations, the presented results here are the lowest motions difference between cases. 135-degree weather heading is selected for all cases as it includes the effects of rolling and pitching together.

Axis system[1]

X – 0 at forward perpendicular, +v towards to aft

Y – 0 at centerline, +v towards to starboard

Z – 0 at keel, +v towards to deck

Direction angles start from 0 coming from aft to 180 degrees coming from bow around starboard side.

Results

The final results are the culmination of an extensive study that involved 36 distinct analyses. These analyses covered a wide range of conditions, encompassing all the weather cases detailed in Table 3 and the various mooring configurations outlined in Table 4. The key findings presented are the maximum motion values recorded for the construction vessel during the time domain analyses.

In addition to the vessel's motion, other critical data points are also included and are presented in Table 8, Table 9 and Table 10 for clarity and detailed review. These include:

- Mooring line tensions
- Soft rope tensions
- Fender comparisons

This comprehensive approach ensures that all aspects of the construction vessel and its associated behaviour are thoroughly documented and analyzed.

Table 6—Motions amplitude results for case initial vs case0, *heave represent the draft level after heave.

Cases		Weather			Motions amplitude (m/deg)					
m. case	w. case	Dir (deg)	Hs(m)	Tp (Sec)	Surge	Sway	Heave*	Roll	Pitch	Yaw
Initial	1	135	1.5	5	0.03	-1.02	-5.25	0.1	-0.15	0.86
Initial	2	135	1.5	6	0.04	-0.88	-5.11	0.31	-0.06	0.95
Initial	3	135	1.5	7	0.07	-0.61	-4.92	0.63	0.07	1.13
case0	1	135	1.5	5	0.01	-1.5	-5.25	0.06	-0.15	1.16
case0	4	135	1.5	6	0.04	-1.37	-5.11	0.25	-0.06	1.26
case0	7	135	1.5	7	0.11	-0.92	-4.91	0.55	0.08	1.38
Initial	4	135	2	5	0.06	-1.05	-5.22	0.13	-0.13	1.23
Initial	5	135	2	6	0.08	-0.81	-5.04	0.4	-0.01	1.4
Initial	6	135	2	7	0.12	-0.33	-4.77	0.8	0.16	1.67
case0	10	135	2	5	0.03	-1.88	-5.21	0.09	-0.13	1.8
case0	13	135	2	6	0.06	-1.72	-5.04	0.34	-0.01	1.94
case0	16	135	2	7	0.14	-1	-4.77	0.71	0.17	2.12
Initial	7	135	2.5	5	0.1	-1.15	-5.18	0.16	-0.11	1.72
Initial	8	135	2.5	6	0.13	-0.8	-4.96	0.49	0.04	1.97
Initial	9	135	2.5	7	0.17	-0.03	-4.63	0.94	0.25	2.32
case0	19	135	2.5	5	0.05	-2.51	-5.18	0.13	-0.11	2.7
case0	22	135	2.5	6	0.08	-2.29	-4.96	0.46	0.04	2.86
case0	25	135	2.5	7	0.17	-1.29	-4.62	0.88	0.26	3.12

Results of Initial cases vs case0

This section highlights the effect of the optimized mooring arrangement on vessel motion for the CV without CB mooring (initial case) and after CB mooring to CV (cas0), for the comparison of intact and damage cases results (case0, case1 and case2) please refer to Appendix-A. While the refined setup slightly increases motion amplitudes, it remains within acceptable limits and supports improved operational efficiency,

Table 6 presents the initial case motion versus case0 motion, Figure 2 and Figure 3 presents the slight increase in sway and yaw as samples of in-plan motion, while Figure 4 presents the roll damping effect due to multi-body mooring.

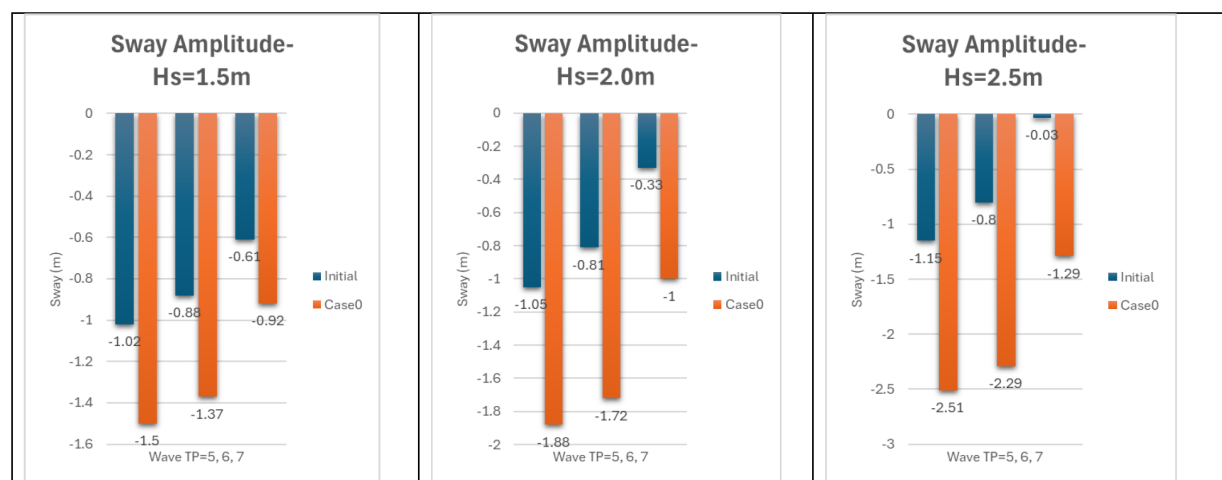


Figure 2—Sway amplitude comparison between initial case and case0 for three Hs values

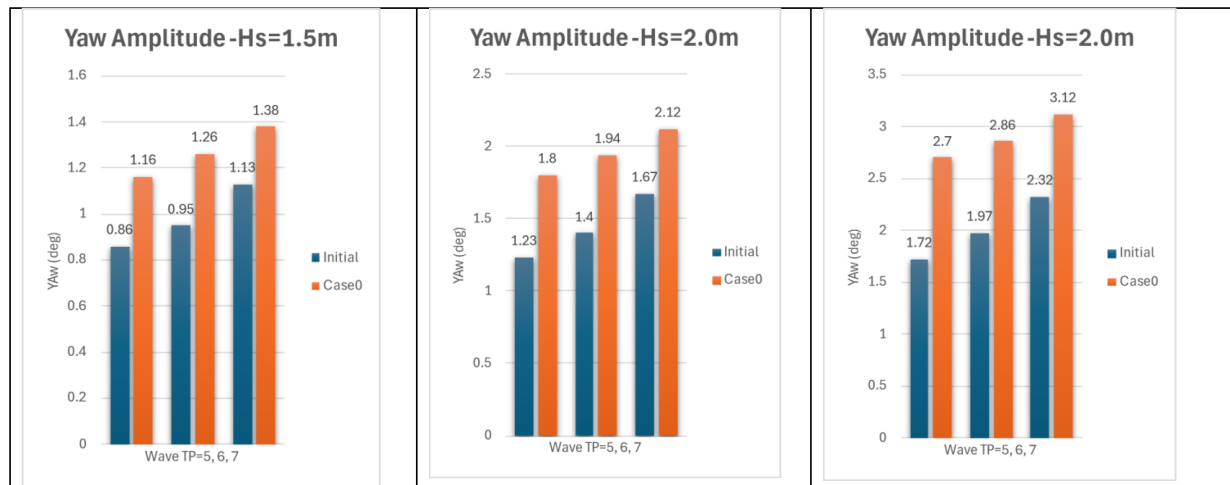


Figure 3—Yaw amplitude comparison between initial case and case0 for three Hs values

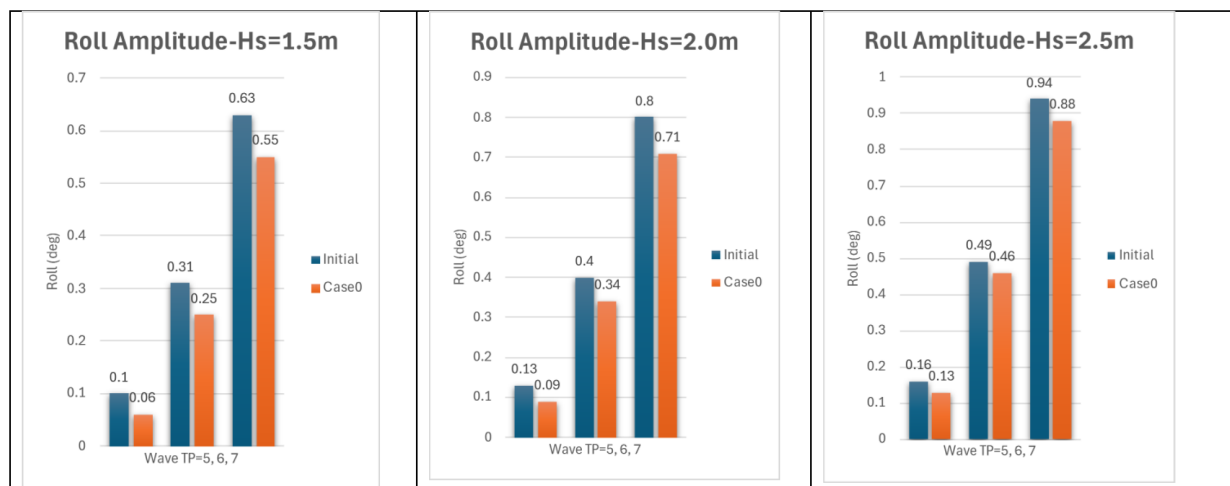


Figure 4—Roll amplitude comparison between initial case and case0 for three Hs values

Conclusion

As a result of the initial and case0 comparison it's noted that the mooring of CB to CV slightly increases the plan motions (surge, sway and yaw), while acting as a damping in roll results. To get the same motion amplitude values of initial case while CB mooring alongside the following recommendations to be studied widely:

1. A weather reduction with small factors to be applied.
2. Selecting another operational peak period with respect to site specific metocean data.
3. Increasing the mooring lines tensions.
4. Optimize the anchor pattern to fit the most probable weather cases and the operation heading.

Case0, case1 and case2 results show that there are no major changes in amplitudes in case of loss of one fender or 2/3 soft ropes, which prove the slacking case of some ropes during actual operation [Figure 5](#), this is due to high mechanical properties efficiency of fenders and soft mooring lines, the assessment should be re-evaluated in case using a hard stiffness connectors values.

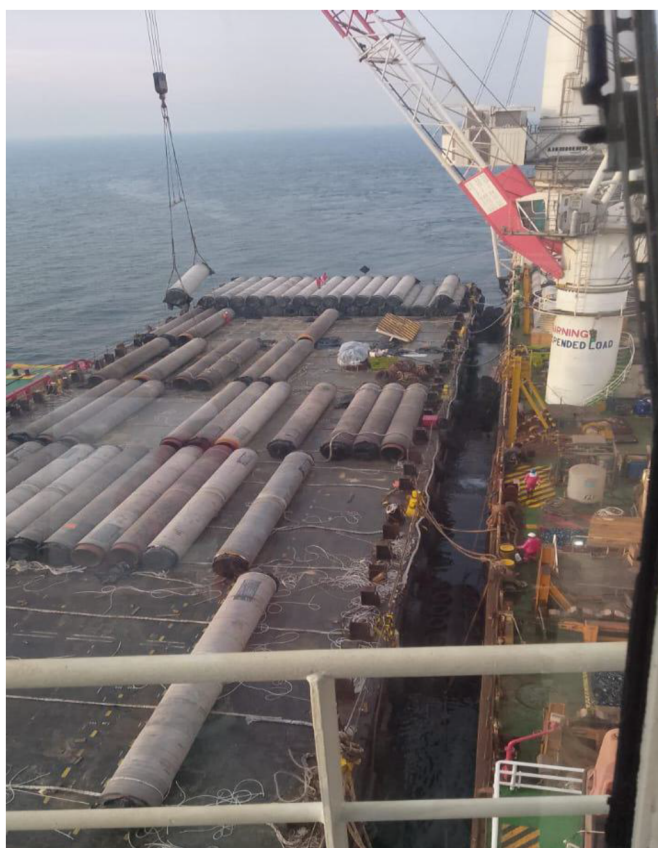


Figure 5—Soft ropes slack case recorded for reference

CB unmooring weather criteria is recommended to be planned and established in the design phase of the marine operation, sufficient towing tug bollard pull is to cover the selected unmooring weather case.

Study Recommendation

The following recommendations are for the ship designer's team during ship design phase:

1. Increasing the bollard pits number and SWL for both units is recommended allowing the design of unit's mooring operation to be optimized.
2. Elevation of bollard pits is recommended to be leveled to cargo barges level, most of large construction vessel depth is triple the level of cargo barges, this results a higher moment to ship's beam and ineffective mooring.
3. Location of fairleads of the vessel is recommended to be at bow and stern, avoiding side fairleads give more space for mooring options.

The following recommendations are for the engineering team during the design of offshore activity:

1. CB equivalent dynamic loads acting on bollards location to be added to the mooring analysis.
2. Lifting analysis and DAF analysis in air should be performed with moored cargo barge.
3. Vessel RAO's phase angle will be a little change with the moored cargo barge, this case to be checked prior laying analysis.

The following recommendations are for the engineering team during establishing the codes and guidelines:

1. Operational weather limit of marine operations to be reduced by multi-body mooring reduction factor, depending on CB displacement, shape and wind area.
2. Alternative approaches to be established for ease of use.
3. Recommendations and guidelines for multi-body mooring to be added to offshore operations guidelines, for the safety of personnel and operations.

The following recommendations are for the Offshore team onboard the construction vessel during mooring and unmooring operation of cargo barge:

1. Multi-body mooring analysis report and mooring arrangement drawing to be reviewed, define the bollard pits with riggers, fenders to be in place as per the drawings, soft ropes to be in place at each selected bollard.
2. Weather window should cover the duration of mooring, offshore activities and unmooring operation.
3. Moore the cargo barge as per the drawing. Towing tug always be ready.
4. Monitor the weather forecast and actual weather conditions with comparing to the limiting weather in the analysis report.
5. In case weather conditions will exceed the limiting weather, offshore activity to be changed to wait on weather, the CB to be unmoored and towing to safe area (anchorage or shelter area).
6. CV team to change the heading and anchor pattern to the wait on weather anchor pattern, work can be continued when the weather reduced again to the limiting weather.

References

1. Bentley Systems, Inc., MOSES Reference Manual, *Moses2024*, Bentley Systems, Inc., 2024.
2. Det Norske Veritas group standard, Marine operations and marine warranty, DNV-ST-N001, 2023.
3. NMDC Group, Safeen3000 specifications, 2024.
4. NMDC Group, Safeen B-281 specifications, 2024.

Appendix-A

Results of Case0, case1 and case2

Motion results in this section show the difference between 3 mooring cases as per [Table 5](#) which describe the effect of mooring configuration to the motions, [Table 7](#) shows the motion results of all mooring cases.

Table 7—Motions amplitude results for case0, case1 and case2, *heave represent the draft level after heave.

Cases		Weather			Motions amplitude (m/deg)					
m. case	w. case	Dir (deg)	Hs(m)	Tp (Sec)	Surge	Sway	Heave*	Roll	Pitch	Yaw
case0	1	135	1.5	5	0.01	-1.5	-5.25	0.06	-0.15	1.16
case1	2	135	1.5	5	0.04	-1.51	-5.25	0.07	-0.15	1.17
case2	3	135	1.5	5	0.01	-1.5	-5.25	0.08	-0.15	1.16
case0	4	135	1.5	6	0.04	-1.37	-5.11	0.25	-0.06	1.26
case1	5	135	1.5	6	0.04	-1.38	-5.11	0.28	-0.06	1.25
case2	6	135	1.5	6	0.03	-1.38	-5.11	0.29	-0.06	1.25
case0	7	135	1.5	7	0.11	-0.92	-4.91	0.55	0.08	1.38
case1	8	135	1.5	7	0.09	-0.93	-4.91	0.59	0.07	1.38
case2	9	135	1.5	7	0.08	-0.94	-4.91	0.59	0.07	1.37
case0	10	135	2	5	0.03	-1.88	-5.21	0.09	-0.13	1.8
case1	11	135	2	5	0.05	-1.87	-5.21	0.11	-0.13	1.82
case2	12	135	2	5	0.02	-1.87	-5.21	0.1	-0.13	1.8
case0	13	135	2	6	0.06	-1.72	-5.04	0.34	-0.01	1.94
case1	14	135	2	6	0.06	-1.72	-5.04	0.38	-0.01	1.95
case2	15	135	2	6	0.06	-1.71	-5.03	0.37	-0.01	1.94
case0	16	135	2	7	0.14	-1	-4.77	0.71	0.17	2.12
case1	17	135	2	7	0.13	-1.01	-4.77	0.77	0.16	2.11
case2	18	135	2	7	0.13	-1.01	-4.77	0.75	0.16	2.09
case0	19	135	2.5	5	0.05	-2.51	-5.18	0.13	-0.11	2.7
case1	20	135	2.5	5	0.03	-2.47	-5.18	0.15	-0.11	2.72
case2	21	135	2.5	5	0.04	-2.49	-5.18	0.12	-0.11	2.69
case0	22	135	2.5	6	0.08	-2.29	-4.96	0.46	0.04	2.86
case1	23	135	2.5	6	0.08	-2.28	-4.96	0.48	0.04	2.87
case2	24	135	2.5	6	0.08	-2.27	-4.96	0.46	0.04	2.85
case0	25	135	2.5	7	0.17	-1.29	-4.62	0.88	0.26	3.12
case1	26	135	2.5	7	0.15	-1.29	-4.63	0.9	0.25	3.1
case2	27	135	2.5	7	0.17	-1.3	-4.62	0.89	0.25	3.07

Table 8—Mooring Lines tensions results for case0, case1, case2

Case	Weather			Mooring Lines tensions (tonne)							
	Dir (deg)	Hs(m)	Tp (Sec)	P1	P2	P3	P4	S1	S2	S3	S4
1	135	1.5	5	27.74	20.47	33.75	28.9	50.22	65.46	45.75	52.16
2	135	1.5	5	27.52	20.2	33.41	29.85	50.47	65.5	46.19	53.16
3	135	1.5	5	27.72	20.43	33.54	28.91	50.31	65.42	45.65	52.16
4	135	1.5	6	29.24	21.93	35.31	29.56	51.93	68.08	49.87	54.15
5	135	1.5	6	28.67	21.77	34.78	30.34	51.72	67.72	49.48	54.43
6	135	1.5	6	28.89	21.81	35.1	29.46	51.64	67.89	49.93	53.85
7	135	1.5	7	32	26.32	39.82	33.25	54.9	72.76	55.4	56.91
8	135	1.5	7	32.23	25.93	38.62	33.55	54.42	72.25	54.09	56.98
9	135	1.5	7	31.54	25.44	39.13	34.79	54.37	72.26	53.63	55.97
10	135	2	5	26.17	17.19	34.37	26.19	57.05	80.58	50.71	63.88
11	135	2	5	25.68	16.76	34.02	28.15	57.2	80.22	51.44	65.71
12	135	2	5	26.26	17.24	34.29	26.26	57.17	80.46	50.82	63.77
13	135	2	6	27.77	19.08	35.88	27.2	59.07	83.79	56.48	67.36
14	135	2	6	27.36	18.75	35.3	28.07	59.18	83.55	56.99	67.48
15	135	2	6	27.62	18.91	35.59	27.09	58.78	83.92	56.99	66.45
16	135	2	7	31.77	25.34	41.07	34.36	62.4	89.66	62.16	71.75
17	135	2	7	32.29	24.88	39.65	34.3	62.78	90.29	61.04	72.09
18	135	2	7	31.3	24.74	40.22	36.48	62.5	89.67	61.47	69.63
19	135	2.5	5	23.88	12.37	34.51	22.98	65.33	100.32	58.16	83.82
20	135	2.5	5	22.99	11.89	33.77	24.49	65.18	99.74	58.08	86.25
21	135	2.5	5	24.03	12.46	34.58	23.03	65.3	100.37	57.65	83.17
22	135	2.5	6	25.77	14.44	35.51	25.27	67.51	104.21	65.25	86.74
23	135	2.5	6	25.27	14.08	35.15	24.69	67.5	103.86	64.95	88.16
24	135	2.5	6	25.93	14.48	35.47	25.02	67.25	104.36	65.6	85.89
25	135	2.5	7	30.33	22.7	40.76	34.46	71.65	112.26	70.12	94.35
26	135	2.5	7	30.65	22.8	39.2	33.15	72.64	113.13	70.48	94.74
27	135	2.5	7	30.54	22.49	40.06	35.88	72.2	112.63	70.25	93.41

Table 9—Fender's compression results for case0, case1, case2

Case	Weather			Fenders Compression (tonne)		
	Dir (deg)	Hs(m)	Tp (Sec)	F1	F2	F3
1	135	1.5	5	7.38	7.01	9.61
2	135	1.5	5	11.31	0	12.26
3	135	1.5	5	10.38	0	12.8
4	135	1.5	6	11.48	8.2	11.28
5	135	1.5	6	13.55	0	13.35
6	135	1.5	6	13.14	0	14.63
7	135	1.5	7	16.36	10.19	13.98
8	135	1.5	7	18.62	0	15.74
9	135	1.5	7	18.49	0	18.07
10	135	2	5	11.51	11.1	14.77
11	135	2	5	18.34	0	19.2
12	135	2	5	17.07	0	19.92
13	135	2	6	16.84	11.79	16.02
14	135	2	6	21.03	0	19.46
15	135	2	6	20.88	0	21.09
16	135	2	7	23.43	14.32	19.17
17	135	2	7	27.57	0	21.83
18	135	2	7	27.69	0	24.52
19	135	2.5	5	17.3	16.5	21.05
20	135	2.5	5	26.21	0	28.12
21	135	2.5	5	25.15	0	29.44
22	135	2.5	6	22.81	16.73	21.55
23	135	2.5	6	30.72	0	27.95
24	135	2.5	6	30.49	0	28.81
25	135	2.5	7	32.47	19.18	26.5
26	135	2.5	7	37.07	0	30.56
27	135	2.5	7	37.05	0	33.8

Table 10—Soft ropes tension results for case0, case1, case2

Case	Weather			Soft ropes tensions (tonne)						
	Dir (deg)	Hs(m)	Tp (Sec)	SR1	SR2	SR3	SR4	SR5	SR6	SR7
1	135	1.5	5	4.35	3.92	5.1	7	12.21	6.58	7.07
2	135	1.5	5	9.34	3.38	0	5.62	10.39	0	5.49
3	135	1.5	5	3.53	0	3.99	0	14.04	4.52	0
4	135	1.5	6	9.41	6.22	9.72	10.05	19.16	13.18	14.87
5	135	1.5	6	9.13	3.63	0	5.86	15.12	0	8.98
6	135	1.5	6	5.05	0	5.26	0	21.83	7.7	0
7	135	1.5	7	15.63	13.96	18.65	18.94	25.26	25.65	28.65
8	135	1.5	7	14.06	5.65	0	8.03	18.33	0	18.75
9	135	1.5	7	6.78	0	8.08	0	29.49	14.6	0
10	135	2	5	5.93	4.18	6.47	7.61	14.44	7.62	6.91
11	135	2	5	18.26	3.81	0	6.8	12.72	0	5.63
12	135	2	5	6.11	0	6.59	0	16.27	7.45	0
13	135	2	6	9.68	6.61	10.14	11.59	22.25	14.3	15.46
14	135	2	6	13.47	4.66	0	8.3	18.83	0	9.48
15	135	2	6	6.64	0	7.25	0	24.64	10.34	0
16	135	2	7	14.6	14.33	18.48	18	28.08	28.94	34.7
17	135	2	7	17.17	7.01	0	8.49	21.03	0	23.68
18	135	2	7	9.75	0	11.91	0	36.4	20.31	0
19	135	2.5	5	8.77	4.43	9.81	8.27	16.71	11.39	6.72
20	135	2.5	5	28.68	4.47	0	7.06	15.5	0	6.6
21	135	2.5	5	9.15	0	9.5	0	18.18	10.45	0
22	135	2.5	6	10.68	7.18	10.94	12.26	24.77	16.3	16.77
23	135	2.5	6	21.1	4.43	0	9.3	21.02	0	11.42
24	135	2.5	6	8.71	0	10.55	0	26.73	13.54	0
25	135	2.5	7	16.35	14.08	21.3	18.65	30.46	34.76	37.07
26	135	2.5	7	21.88	8.94	0	9.32	21.18	0	27.67
27	135	2.5	7	11.88	0	14.66	0	39.45	26.39	0